SOKOINE UNIVERSITY OF AGRICULTURE

Department of Agricultural Engineering

AGE 214

INTRODUCTION TO ELECTRONICS

COMPENDIUM OF LECTURE NOTES

LECTURE 1

Introduction to Electronics & Semiconductor Physics

Lab Facility:

Electronics and Precision Agriculture Lab

Instructor:

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Chapter 1

Introduction to Electronics & Semiconductor Physics

1.1 Course Overview and Objectives

Welcome to **AGE 214: Introduction to Electronics**. This foundational course is designed to equip agricultural engineering students with the essential knowledge of electronic components and circuits. As agriculture transitions into the digital age (Precision Agriculture), the ability to understand, design, and troubleshoot electronic systems is becoming as critical as understanding mechanics or hydraulics.

The course is managed by the **Electronics and Precision Agriculture Lab**. Throughout this semester, we will journey from the sub-atomic physics of semiconductors to the practical design of amplifiers and power supplies.

1.1.1 Learning Outcomes

By the end of this lecture series, you should be able to:

- Define electronics and distinguish it from electrical power engineering.
- Explain the atomic structure of materials using the Bohr Model.
- Differentiate between Conductors, Insulators, and Semiconductors based on Energy Band Theory.

- Describe the crystal lattice structure of Silicon and the nature of covalent bonding.
- Understand the concept of ”Holes” and ”Free Electrons” as charge carriers.

1.2 What is Electronics?

Electronics is defined as the branch of science and engineering that deals with the emission, flow, and control of electrons in vacuum, gases, and semiconductors.

It is important to distinguish this from **Electrical Engineering**. While the boundaries often overlap, the primary distinction lies in the application:

- **Electrical Engineering** typically focuses on the generation, transmission, and distribution of electrical power (High Voltage, High Current). Examples include motors, generators, and the national grid.
- **Electronics** focuses on the control of electron flow to process information or control energy (Low Voltage, Micro-currents). Examples include sensors, microprocessors, and amplifiers.

1.3 The Structure of Matter

To understand how a semiconductor works—and ultimately how a transistor amplifies a signal—we must look at the atomic level. All matter is composed of atoms.

1.3.1 The Bohr Atomic Model

Proposed by Neils Bohr in 1913, this model describes the atom as a miniature solar system:

1. **Nucleus:** The central core containing Protons (Positively charged) and Neutrons (Neutral). It contains almost all the mass of the atom.
2. **Electrons:** Negatively charged particles that orbit the nucleus in specific paths called shells or energy levels.

1.3.2 Electron Shells

Electrons do not orbit randomly. They occupy specific discrete energy levels ($n = 1, 2, 3...$). The maximum number of electrons (N_{max}) that can exist in a shell is determined by the formula:

$$N_{max} = 2n^2 \tag{1.1}$$

Where n is the shell number (Principal Quantum Number).

- **K Shell ($n = 1$):** $2(1)^2 = 2$ electrons.
- **L Shell ($n = 2$):** $2(2)^2 = 8$ electrons.
- **M Shell ($n = 3$):** $2(3)^2 = 18$ electrons.

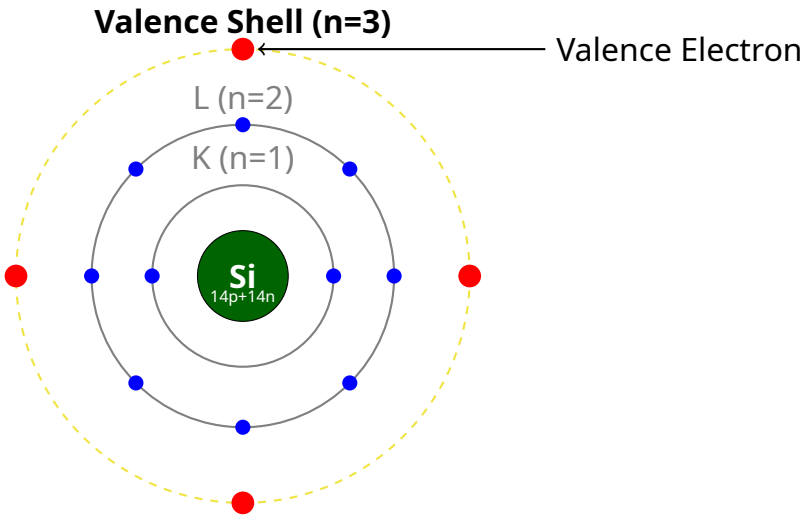


Figure 1.1: Bohr Model of a Silicon Atom (Atomic Number 14). Note the 4 Valence Electrons.

1.3.3 Valence Electrons

The electrons in the outermost shell are known as **Valence Electrons**. These are the most critical electrons in electronics because:

- They are loosely bound to the nucleus.
- They determine the electrical conductivity of the material.
- They participate in chemical bonding.

- An atom with 1 valence electron is an excellent conductor (e.g., Copper).
- An atom with 8 valence electrons is an insulator (Inert gas).
- An atom with 4 valence electrons is a **Semiconductor** (e.g., Silicon).

1.4 Energy Band Theory

In a single isolated atom, electrons occupy discrete energy levels. However, when atoms are packed together to form a solid crystal lattice, these discrete levels split into bands of allowed energies.

1.4.1 The Energy Bands

1. **Valence Band:** The lower energy band containing the valence electrons. These electrons are attached to the atom and do not contribute to current.
2. **Conduction Band:** The higher energy band. If an electron gains enough energy (from heat or light) to jump here, it becomes a ”Free Electron” and can move through the lattice, creating current.
3. **Forbidden Energy Gap (E_g):** The gap between the valence and conduction bands. No electrons can exist here. An electron must gain energy equal to E_g to jump across.

1.5 Classification of Materials

Based on the Energy Band Theory, we classify materials into three categories:

1.5.1 Insulators

Insulators are materials that prevent current flow.

- **Structure:** Valence electrons are tightly bound.
- **Energy Gap:** Very wide ($E_g > 5$ eV).

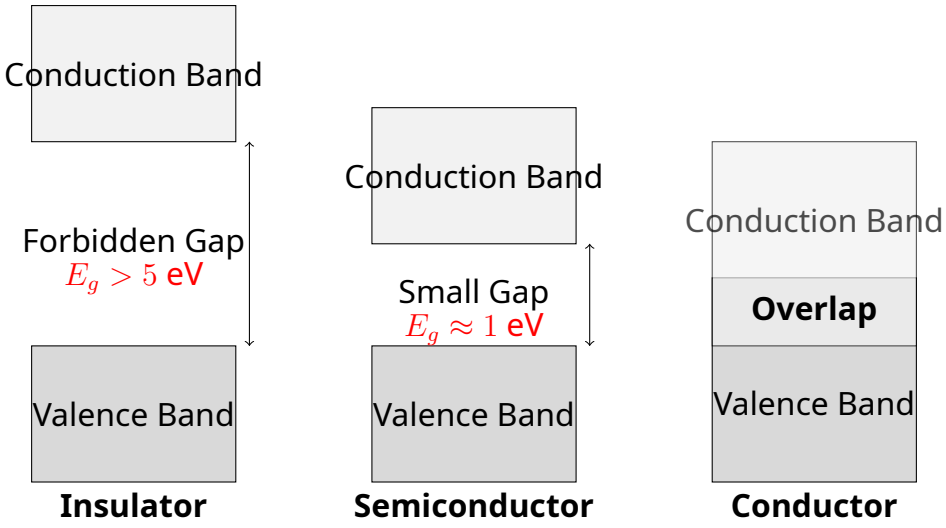


Figure 1.2: Energy Band Diagrams distinguishing Insulators, Semiconductors, and Conductors.

- **Behavior:** At room temperature, thermal energy is insufficient to lift electrons into the conduction band.
- **Examples:** Glass, Rubber, Ceramic, Diamond ($E_g \approx 6$ eV).

1.5.2 Conductors

Conductors allow current to flow with minimal resistance.

- **Structure:** Valence electrons are very loosely bound (Electron Sea).
- **Energy Gap:** Zero ($E_g \approx 0$ eV). The Valence and Conduction bands overlap.
- **Behavior:** Electrons flow freely even with very little applied energy.
- **Examples:** Copper (Cu), Silver (Ag), Gold (Au).

1.5.3 Semiconductors

Semiconductors have conductivity between conductors and insulators.

- **Structure:** They are **Tetravalent** (4 valence electrons).
- **Energy Gap:** Small ($E_g \approx 1$ eV).
- **Behavior:**

- At 0 K (Absolute Zero): They act as perfect insulators.
- At 300 K (Room Temperature): Thermal energy breaks some bonds, allowing some electrons to jump to the conduction band.

- **Examples:** Silicon (Si), Germanium (Ge).

1.6 Semiconductor Materials: Silicon vs. Germanium

1.6.1 Silicon (Si)

- **Atomic Number:** 14.
- **Band Gap (E_g):** 1.1 eV at room temperature.
- **Pros:** Abundant (sand), cheap, high temperature tolerance ($200°$C). Forms perfect oxide (SiO_2) for ICs.

1.6.2 Germanium (Ge)

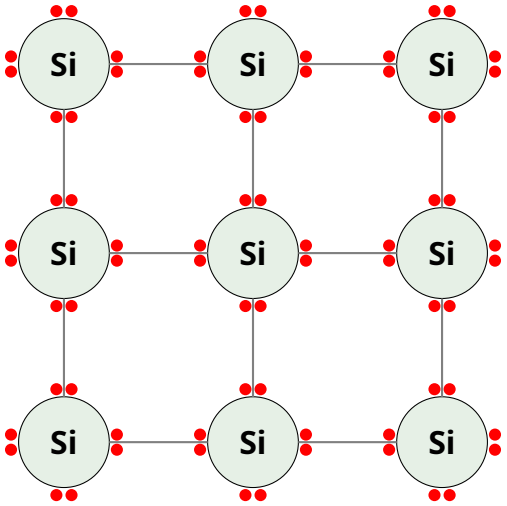
- **Atomic Number:** 32.
- **Band Gap (E_g):** 0.67 eV.
- **Cons:** Sensitive to heat (unstable above $75°$C), high leakage current. Rarely used today.

1.7 Covalent Bonding in Semiconductors

An **Intrinsic Semiconductor** is a pure crystal with no impurities. Silicon atoms hold together by sharing electrons.

1.7.1 The Crystal Lattice

Each Silicon atom shares its 4 valence electrons with 4 neighboring atoms. This sharing forms **Covalent Bonds**. The result is a stable crystal structure where every atom effectively has 8 electrons in its outer shell (a stable octet).



Each Si atom shares 4 electrons with neighbors

Figure 1.3: 2D Representation of Intrinsic Silicon Lattice Structure at 0K.

1.7.2 Thermal Generation

At absolute zero, all electrons are locked in these covalent bonds. However, as temperature rises:

1. Heat energy causes the atoms to vibrate.
2. Occasionally, a covalent bond breaks.
3. An electron is dislodged and becomes a **Free Electron** (Negative Carrier).
4. The vacancy left behind in the bond is called a **Hole** (Positive Carrier).

1.8 Current Flow: Electrons and Holes

In metallic conductors, current is carried only by free electrons. In semiconductors, current is the result of the movement of **both** electrons and holes.

1.8.1 Electron Current

When a voltage is applied, free electrons in the conduction band move toward the positive terminal.

1.8.2 Hole Current

This is a concept unique to semiconductors.

- A hole behaves like a positive charge.
- A valence electron from a neighboring bond moves to fill a hole.
- This leaves a new hole in the neighbor's position.
- As electrons jump to fill holes, the "Hole" appears to move in the opposite direction (towards the negative terminal).

Intrinsic Carrier Concentration

In a pure semiconductor, electrons and holes are generated in pairs. Therefore:

$$n = p = n_i$$

Where:

- n = Concentration of free electrons
- p = Concentration of holes
- n_i = Intrinsic carrier concentration (function of temperature)

1.9 Review Questions

1. Define "Electronics" and explain how it differs from "Electrical Engineering".
2. With the aid of a diagram, explain the difference between Insulators, Semiconductors, and Conductors using Energy Band Theory.
3. Why does Silicon have a higher resistance at 0 Kelvin compared to Room Temperature?
4. Explain the concept of "Hole Current". Does a hole physically move?
5. Calculate the maximum number of electrons that can occupy the M shell ($n = 3$).